Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
(University under section 2(f) of UGC Act, 1956)

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech, ME III semester
Name of the Paper: Production Technology
Paper Code: TME-309

Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. What role does technological advancement play in maintaining competitive advantage in the manufacturing sector? (CO1)

OR

b. Define the different types of patterns and allowances used in casting. (CO1)

Q2. (10 Marks)

a. What do you understand by casting? Write down the properties of molding sand. (CO1)

OR

b. What are the different parts of gating system? Explain with a neat sketch. (CO1)

Q3. (10 Marks)

a. What are the different defects in casting? Also write their cause and remedies. (CO2)

OR

b. What will be the solidification time for a 1100 mm diameter and 33 mm thick casting of Aluminium if the mould constant is 2.2 sec/mm^2? (CO1)

Q4.

a. Write down the difference between Hot working and Cold working. (CO2)

OR

b. What are the different types of casting processes? Define. (CO2)

Q5. (10 Marks)

a. Write short notes on yield criteria for ductile materials and plastic stress strain relationship. (CO2)

OR

b. Write down the classification of metal working processes. (CO2)

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